Acidic extraction of histone proteins

Histones are highly basic proteins that can be selectively extracted from cells, nuclei, or chromatin using dilute acids—a method first described by Kossel in 1884. This modernized protocol, adapted from Holt et al. (2021), enables efficient recovery of histones from small or precious samples.
The protocol uses acid extraction followed by trichloroacetic acid (TCA) precipitation to isolate total histones while minimizing loss of many post-translational modifications. It is particularly suited for downstream applications including Western blotting, immunoprecipitation, and LC-MS/MS proteomics.

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This is a bench card. Full protocol available online.

Procedures

›› *Acid extraction*

- ☐ 0.2 M Hydrochloric acid, 1 mL
- ☐ 100% Trichloroacetic acid
- ☐ 99.9% Acetone, 1 mL
- ☐ 100% Acetone

(1.) Start with a nuclei pellet or a nuclear fraction isolated from approximately 1.0×10^{6}–1.0×10^{7} cells. Approximate the packed cell volume; this is 1 vol.

(2.) Gently vortex the pellet from the preceding nuclear isolation or extraction procedure.

(3.) *Critical:* In a dropwise manner, add 5 vol (10 vol if starting with fewer than 1×10^{6} cells) of 0.2 M hydrochloric acid to the pellet at 4 °C. Gently pipette up and down to disperse the pellet.

> ***Critical:*** Add slowly to avoid rapid precipitation. If working with small volumes, pipette gently rather than dropwise.

(4.) Incubate on ice for 5 min. 5 min

(5.) Centrifuge at 15 000 × *g* for 5 min at 4 °C. Transfer the supernatant to a clean tube.

(6.) To the supernatant, add 0.25 vol of 100% trichloroacetic acid. Vortex vigorously to precipitate histones.

> ***Critical:*** Mix immediately and completely to avoid incomplete precipitation.

(7.) Precipitate on ice for 5 min. 5 min

(8.) Collect the histone pellet at 15 000 × *g* for 5 min at 4 °C. Discard supernatant.

(9.) Carefully add 100 µL cold (−20 °C) 99.9% acetone with HCl to the pellet. Do not disturb the pellet!

> ***Quality assurance:*** Since acetone can degrade some plastics, perform washes quickly in compatible tubes.

(10.) Centrifuge at 15 000 × *g* for 2 min at 4 °C. Remove acetone using a glass pipette.

> ***Quality assurance:*** Do not scrape the walls. Leave residual solvent rather than risk losing pellet material. A small amount of acetone will evaporate quickly.

(11.) Repeat acetone wash using 100 µL pure acetone (without HCl).

(12.) Air dry the pellet at room temperature.

> ***Critical:*** Avoid overdrying. Desiccated pellets may be difficult to redissolve.

Storage of isolated histones

(1.) Store the histone pellet at room temperature for up to one week. For long-term storage, keep at −80 °C. 🕮

List of references

M.V. Holt, T. Wang, and N.L. Young, *Curr. Protocol.* **1**(2), e26 (2021).

A. Kossel, *Hoppe-Seyler's Z. Physiol. Chem.* **8**(6), 511–515 (1884).

Recipe (available online) 🕮 Notes (available online)